

LESSON PLAN

Faculty Name: Dr. S. Meenakshi	Department: Computer Science and Engineering
Subject: Computer Networks (CS501)	Semester: V
Academic Year: 2025–2026	Unit: III – Network Layer
Topic: Routing Algorithms – Comparing Dijkstra and Bellman-Ford	Duration: 90 Minutes
Date: 30 April 2026	

Course Outcome: CO3 – Analyze and compare network routing algorithms with respect to convergence, complexity, and failure scenarios.

Program Outcome: PO2 – Problem Analysis, PO3 – Design Solutions

1. LEARNING OBJECTIVES

By the end of this lesson, students will be able to:

- Differentiate between Distance Vector and Link State routing paradigms.
- Analyze the step-by-step execution of Dijkstra's algorithm on a given network graph.
- Examine the Bellman-Ford algorithm and trace it through a sample topology.
- Compare the two algorithms with respect to time complexity and convergence speed.

Bloom's Level: Analyze (a–d)

2. PREREQUISITE KNOWLEDGE

- Graph representations (adjacency matrix, adjacency list) from Data Structures course.
- OSI model and Network layer role (Unit I).

3. CONTENT OUTLINE

3.1 Routing Overview (15 min)

- Goal of dynamic routing algorithms: Link State (LS) vs Distance Vector (DV).

3.2 Dijkstra's Algorithm – Link State Routing (25 min)

- Iterative shortest path computation using global topology state.

3.3 Bellman-Ford Algorithm – Distance Vector Routing (25 min)

- Local neighborhood cost exchanges using the decentralized Bellman-Ford update equation.

3.4 Routing Inquiry Activity (25 min)

- Students trace step-by-step iterations on a 6-node network graph worksheet.

4. TEACHING METHODS AND STRATEGIES

- Method 1: Lecture and Whiteboard worked examples to trace execution routing tables.
- Method 2: Activity-Based / Small group collaboration worksheets tracing graph weights.

5. ASSESSMENT STRATEGY

Formative:

- In-class worksheet review tracking Dijkstra iteration step tables.

Summative:

- Network Routing Lab Exam: Students trace topology link failure responses. Marks: 30.

6. RESOURCES AND MATERIALS

- Textbook: Kurose & Ross, "Computer Networking: A Top-Down Approach," 8th Ed.

